

COPPER MOUNTAIN, CO, USA

WMO: 722061

Lat: 39.4752N

Lon: 106.1520W

Elev: 12074

StdP: 9.32

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 03038

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-6.1	-1.6	-29.5	1.7	24.6	-21.9	2.6	22.4	37.5	13.2	34.6	14.9	13.6	270	0.908

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	68.0	43.7	64.3	43.0	63.5	42.9	49.2	59.3	47.9	57.9	46.6	57.4	11.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
45.1	70.4	54.1	43.2	65.3	52.9	41.2	60.5	52.3	24.4	58.6	23.5	58.1	22.7	57.6	54.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.5	30.5	26.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.3	17.2	18.0	24.3	28.6	37.4	49.9	56.6	53.1	47.3	35.0	26.3	16.6
	DBStd	16.11	8.74	8.53	8.77	8.26	8.03	7.93	4.51	4.96	7.17	9.24	8.97	9.43
	HDD50	6212	1018	896	796	643	394	96	7	26	123	466	712	1035
	HDD65	11216	1483	1316	1261	1093	857	454	261	369	530	930	1162	1500
	CDD50	473	0	0	0	0	3	92	211	122	44	1	0	0
	CDD65	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDH74	2	0	0	0	0	0	1	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	13.1	13.6	15.3	14.7	15.0	13.3	13.1	10.5	10.7	11.6	13.6	13.2	13.0
Precipitation	PrecAvg	17.9	1.2	1.2	1.5	1.9	1.4	1.1	2.2	2.4	1.7	1.2	1.2	1.0
	PrecMax	24.2	3.9	3.0	2.4	3.0	4.1	2.2	4.3	3.7	2.2	2.1	1.9	3.1
	PrecMin	12.4	0.2	0.4	0.6	0.6	0.3	0.3	0.5	0.9	1.0	0.4	0.5	0.0
	PrecStd	2.9	0.9	0.7	0.6	0.6	0.8	0.6	1.1	0.8	0.4	0.5	0.4	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.5	37.4	46.0	51.6	59.3	72.0	72.3	68.2	67.5	58.5	46.2	37.3
		MCWB	25.4	26.3	29.8	33.8	36.7	42.8	46.1	44.2	40.5	37.3	31.0	23.9
	2%	DB	34.1	35.8	41.5	45.4	54.4	65.8	68.4	64.2	63.0	54.0	43.0	34.4
		MCWB	24.3	25.2	27.7	30.4	35.9	40.6	44.6	43.8	40.9	36.3	28.6	23.0
	5%	DB	31.7	33.6	37.4	43.2	52.4	63.7	65.9	63.3	59.4	49.9	40.5	32.4
		MCWB	23.7	24.5	26.3	29.2	35.2	40.2	44.3	43.6	39.9	34.5	27.7	22.9
	10%	DB	28.3	29.9	36.1	39.5	47.8	60.9	63.8	60.9	58.6	45.7	36.8	29.7
		MCWB	21.7	23.1	25.7	27.4	33.3	39.3	44.0	42.9	39.8	32.6	26.2	23.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.7	30.1	32.2	34.8	39.7	46.4	51.3	50.9	46.5	42.0	33.5	28.1
		MCDB	34.5	36.4	40.9	44.1	52.4	60.9	61.3	59.5	57.1	50.6	40.4	32.7
	2%	WB	26.3	27.0	29.9	32.8	37.8	44.6	49.7	48.9	44.4	38.9	31.4	26.0
		MCDB	31.8	32.0	39.5	41.1	50.5	59.1	60.1	58.2	56.5	48.7	38.4	31.4
	5%	WB	24.2	25.0	28.3	31.4	36.3	43.2	48.5	47.5	43.0	36.6	29.5	24.0
		MCDB	29.3	30.6	37.2	39.5	47.9	58.0	58.9	56.6	55.4	46.0	37.6	29.7
	10%	WB	22.8	23.4	26.4	29.9	35.0	41.8	47.5	46.1	41.6	34.7	27.9	22.8
		MCDB	27.7	29.2	34.4	38.3	45.7	56.7	57.9	56.3	53.4	43.8	36.1	28.5
Mean Daily Temperature Range	5% DB	MDBR	11.5	11.6	12.7	12.8	12.7	14.6	15.3	14.4	15.1	12.9	11.2	11.1
		MCDBR	12.4	12.6	13.5	13.5	15.3	15.9	16.6	15.4	16.2	16.0	12.9	10.7
		MCWBR	8.3	8.3	7.9	7.2	7.2	7.1	6.9	6.8	7.2	7.7	6.9	7.3
	5% WB	MCDBR	11.7	10.9	12.7	12.6	14.0	15.1	14.9	14.3	14.7	14.4	12.0	10.2
		MCWBR	9.1	8.2	8.2	8.4	7.2	7.2	7.4	7.5	7.2	7.2	7.5	8.4
Clear-Sky Solar Irradiance	taub	0.187	0.192	0.211	0.234	0.246	0.252	0.274	0.271	0.240	0.221	0.198	0.188	
	taud	2.532	2.493	2.469	2.530	2.649	2.656	2.633	2.613	2.734	2.743	2.687	2.565	
	Ebn at Noon	331	339	339	333	328	324	316	316	324	325	325	323	
	Edh at Noon	22	27	31	32	29	29	29	29	24	21	19	20	
All-Sky Solar Radiation	RadAvg	864	1183	1613	1823	1946	2283	2034	1797	1627	1207	863	712	
	RadStd	39	60	122	125	152	137	109	92	110	92	70	52	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+124	+38

Nomenclature: See separate page